HAOBIN (HIROKI) CHEN

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EDUCATION

Indiana University Bloomington *Aug. 2023 – May 2028 (Expected)*
Ph.D. in Computer Science
Advisors: Haixu Tang, Chenghong Wang (IU), and XiaoFeng Wang (Nanyang Technological University)

Indiana University Bloomington *Aug. 2023 – Jul. 2026*
M.S. in Computer Science (conferred July 31, 2026)

Nankai University, Tianjin, China *Aug. 2019 – Jun. 2023*
B.Eng. in Information Security

RESEARCH INTERESTS

Confidential computing, systems security, and formal verification.

INDUSTRY EXPERIENCE

Ant Group (Ant Research) May 2026 – Aug. 2026
Research Intern *Beijing, China*

· Extended the Asterinas kernel verification codebase (∼29K LoC) with support for RCU and the weak memory model.
· Implemented concurrent verification of the memory-management (MM) subsystem using an Iris-style resource algebra.

CertiK (Certified Kernel Tech LLC) Oct. 2025 – Mar. 2026
Security Research Intern (Mentors: Dr. Hao Chen, Dr. Sean N. Anderson) *Remote*

· Conducted formal verification of Solana smart contracts and protocols to provide mathematical assurance for digital assets.
· Applied **Verus** to verify the memory-management module of the **Asterinas** Rust OS kernel (USENIX ATC'25), ensuring spatial and temporal memory safety.

TikTok Privacy Innovation Lab (ByteDance Inc.) May 2024 – Aug. 2024
Research Intern (Mentor: Dr. Mingshen Sun) *San Jose, CA*

· Proposed TAVERNS, a decentralized remote-attestation design that reduces reliance on centralized verification services.
· Developed a TEE verification method based on **zero-knowledge proofs**, resulting in U.S. Patent Application Pub. No. US 2026/0106759 A1 (pending).
· Engineered TEE-based data clean rooms (ManaTEE, now a *Confidential Computing Consortium* project) for TikTok's research platform on Google Cloud confidential instances using Kubernetes.

Google Summer of Code: Apache Teaclave Jun. 2023 – Nov. 2023
Open Source Contributor (Mentor: Dr. Mingshen Sun) *Remote*

· Developed data-use-policy definition and enforcement for TEE-based data analysis.
· Integrated formal verification components to strengthen the framework's security guarantees.

RESEARCH EXPERIENCE

DEKO: Verifiable Data Governance in Confidential VMs Jul. 2026 – Present
Research Assistant (with Prof. Chenghong Wang, Dr. Mingshen Sun, and Ben Sternthal); funded by a 2026 CCC Academic Research Grant (PI: C. Wang) *IU*

· Designed intra-CVM isolation that protects trusted governance logic from an untrusted guest OS.
· Built a Verus-verified information-flow-control (IFC) monitor for policy enforcement and provenance tracking.
· Bound provenance and policy events to hardware attestation for cloud-native workloads.

Privacy-Preserving Data Sharing and Analytics (PPDSA) Aug. 2024 – Present
Research Assistant (with Prof. XiaoFeng Wang, Prof. Chenghong Wang, and Dr. Marcela Melara of Intel) *IU*

· Applied confidential computing to secure large-scale data analytics.
· Developed machine-checked verification and policy enforcement for data lakehouses.
· Explored hardware-software co-designs for confidential data analytics.

Center for Distributed Confidential Computing (CDCC) Aug. 2023 – Present
Research Assistant (with Prof. Haixu Tang and Prof. XiaoFeng Wang) *IU*

· Architected hardware-software co-designs for policy-enforced isolation within TEEs.
· Used **Verus** and **Coq** to verify large codebases and security-critical system kernels.
· Optimized the performance of secure systems on CPU and GPU TEEs in cloud and edge environments.

Proof of Being Forgotten: Rust-SGX Enclave Framework May 2022 – Jun. 2023
Research Assistant (with Prof. XiaoFeng Wang and Dr. Mingshen Sun) *Remote*

· Implemented secret-residue cleaning algorithms and custom allocators for Intel SGX in **Rust**.
· Formally verified execution models in **Coq** to ensure that no secrets leak to unauthorized parties.

PUBLICATIONS

- H. Wang, **Haobin Hiroki Chen**, H. Chen, W. Huang, G. Mortensen, Y. Chen, J. Su, H. Tang. A Confidential Computing Framework for Runtime Data Policy Enforcement in Biomedical Data Analytics: Design, Implementation, and Evaluation. In *Proceedings of the 60th Hawaii International Conference on System Sciences* (**HICSS-60**), 2027. To appear.
- Y. Guo, H. Chen, **Haobin Hiroki Chen**, Y. Luo, X. Wang, C. Wang. BOLT: Bandwidth-Optimized Lightning-Fast Oblivious Map Powered by Secure HBM Accelerators. In *Proceedings of the 2025 ACM SIGSAC Conference on Computer and Communications Security* (**CCS '25**), 2025.
- **Haobin Hiroki Chen**, H. Chen, M. Sun, C. Wang, X. Wang. PICACHV: Formally Verified Data Use Policy Enforcement for Secure Data Analytics. In *Proceedings of the 34th USENIX Security Symposium* (**USENIX Security '25**), 2025.
- **Haobin Chen**, Y. Yang, S. Lv. Revisiting Frequency-Smoothing Encryption: New Security Definitions and Efficient Construction. *Cybersecurity* 7, 15 (2024).
- H. Chen, **Haobin Hiroki Chen**, M. Sun, K. Li, Z. Chen, X. Wang. A Verified Confidential Computing as a Service Framework for Privacy Preservation. In *Proceedings of the 32nd USENIX Security Symposium* (**USENIX Security '23**), 2023.

MANUSCRIPTS

- **Haobin Hiroki Chen**, [redacted] Under submission.
- **Haobin Hiroki Chen**. Xinyi Wan, Yuwei Liu , Sean Anderson, and Lin Huang. Towards Verified RCU Reclamation in Asterinas under Relaxed Memory. **Link**

PATENTS

- M. Sun, **Haobin Chen**, D. Lu, Q. Yan. Trustless Execution Environment Cryptographic Proofs. U.S. Patent Application Pub. No. US 2026/0106759 A1, published Apr. 16, 2026 (pending).

TALKS

2026 **Confidential Computing Consortium, Technical Advisory Council** (Sept. 17, 2026): DEKO: Verifiable Data Governance in Confidential VMs.

2025 **Intel Corporation**: PICACHV: Formally Verified Data Use Policy Enforcement for Secure Data Analytics.

HONORS AND AWARDS

2024 **ACM CCS Distinguished Artifact Reviewer**

2023 **Nankai University Distinguished Bachelor Thesis Award** (top 3%)

2021 **Third Prize, National College Student Information Security Contest** (top 8%)

2021–22 **Nankai University Academic Excellence Scholarship** (top 5%)

ACADEMIC SERVICE

Artifact Evaluation Committees: ACM CCS '24; USENIX Security '25; NDSS '26 and '27
Journal Reviewer: IEEE Transactions on Information Forensics and Security; PeerJ Computer Science

TECHNICAL SKILLS

Programming	Rust, C/C++, Python, OCaml, Haskell
Verification	Verus, Coq/Rocq, Lean 4